



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Jewel Digger & Finder

Pitch-Aware Resonant Shimmer

A scale-aligned filter bank finds the musical resonances inside a sound and feeds only those into an octave-up crystal grain loop.



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01 Overview

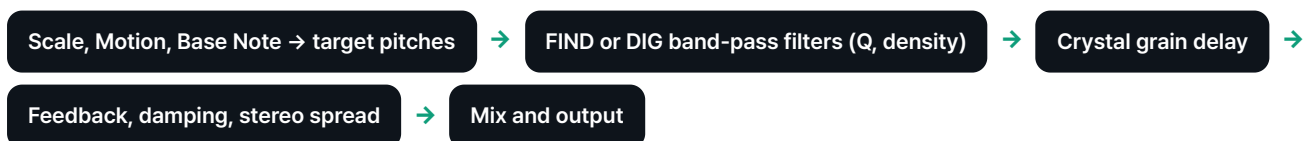
TB Jewel Digger & Finder picks out the harmonic "jewels" in a sound. A filter bank tuned to a base note and scale isolates musical resonances, a motion engine moves them in time, and a crystal grain loop turns them into a shimmering layer, an octave up by default.

Unlike a reverb that washes over everything, the shimmer grows only from the notes you select, so it stays in key with the music. Use it on pads, guitars, piano, vocal tails, field recordings, and ambience.

Key features

- Scale-aligned band-pass filters: base note C0–C8, ten scales, density up to 24 bands
- Two filter modes: FIND lifts the chosen bands, DIG carves into them
- Six Shovel materials that change the resonant character
- Ten motion types, free rate or tempo-synced divisions, motion depth, and randomness
- Crystal grain loop with size, pitch, envelope, delay, feedback, damping, spread, and reverse chance
- Crystal Only mode, low cut, mix, and output
- Display of the filter response on a note scale

Signal flow



About this manual

This manual describes TB Jewel Digger & Finder 1.3.0 for Windows (VST3). The current macOS build is version 1.0.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert the plug-in on a pad, guitar, piano, or ambience track.
- 2 Set **Base Note** 3 and **Scale** 4 to the key of the music.
- 3 Keep **Amount** and **Q** 6 low at first, then raise them until the chosen notes stand out.

- 4 Choose a **Motion** type and set **Rate** 7; start slow.
- 5 Raise **Crystal Mix** and **Crystal Send** in CRYSTALLIZING 9 to grow the shimmer layer.
- 6 Balance the result with **Mix** and **Output** 11.

03 Interface



- 1 **Preset bar** — the preset list, Save, Load, the current filter mode, Undo, and Redo.
- 2 **Response display** — the filter response on a note scale (C0–C8), the base note, and the current motion offset (Move).
- 3 **Base Note** — the root of the filter bank.
- 4 **Scale, Shovel, Motion** — which notes are used, the material character, and how the target moves.
- 5 **MODE** — Find or Dig.
- 6 **Amount, Q, Density** — filter strength, resonance, and number of bands.
- 7 **Rate, Sync, Motion Depth, Random** — speed and range of the motion.
- 8 **Low Cut** — keeps the low end out of the processed signal.
- 9 **CRYSTALLIZING** — the crystal grain loop.
- 10 **Crystal Only** — hear only the crystal layer, without the filter path.
- 11 **MIX / OUTPUT** — final blend and level.

04 Filtering

The filter bank places band-pass filters on the notes of the chosen scale, starting from Base Note. FIND lifts those bands so the chosen notes ring out; DIG cuts into them for a carved, inverted character.

Control	Range · default	What it does
Base Note	C0 to C8 (MIDI 12–108) · A3	The root note of the filter bank.
Scale	Chromatic, Minor, Major, Minor 7, Major 7, Augmented, Diminished, Pentatonic Minor, Pentatonic Major, Mixolydian · Chromatic	Which notes receive a filter band.
Shovel	Crystal, Hollow, Bow, Hammer, Metal, Dirt · Crystal	The material character of the resonance.
MODE Find / Dig	Find, Dig · Find	Find emphasizes the selected bands; Dig cuts into them.
Amount	0 to 100% · 35%	Strength of the Find or Dig filtering.
Q	0.35 to 24 · 6	Resonance of each band. High values ring like tuned strings.
Density	1 to 24 · 12	Number of filter bands.

Motion

Control	Range · default	What it does
Motion	Still, Arp Up, Arp Down, Ramp Up, Ramp Down, Sine, Square, Triangle, Musical Random, Chaos · Arp Up	How the target pitches move: not at all, stepping through the scale, sweeping, or randomly.
Rate	0.02 to 64 Hz · 0.5 Hz	Motion speed when Sync is off.
Sync	Off, On · Off	Locks the motion to the host tempo using Division.
Division	1/64 to 1/1 · 1/4	Note value for synced motion.
Motion Depth	0 to 24 st · 7 st	How far the targets move, in semitones.
Random	0 to 100% · 50%	Amount of randomness. It appears dimmed when the selected Motion does not use it.

Synced motion follows the host tempo between 20 and 320 BPM.

05 Crystallizing

The crystal stage is a grain delay loop that receives the filtered notes, pitches them (an octave up by default), and feeds them back to build a shimmering layer above the source.

Control	Range · default	What it does
Crystal Send	0 to 100% · 80%	How much of the filtered signal enters the crystal loop.
Crystal Mix	0 to 100% · 35%	Level of the crystal layer in the output.
Pitch	−24 to +24 st · +12 st	Pitch of the grains. +12 st is one octave up.
Size	15 to 400 ms · 120 ms	Grain length.
Attack	1 to 75% of Size · 35%	Grain fade-in, shown in milliseconds (42 ms at the default Size).
Decay	1 to 95% of Size · 65%	Grain fade-out, shown in milliseconds (78 ms at the default Size).
Delay	0 to 1000 ms · 140 ms	Delay before the grains repeat.
Feedback	0 to 100% · 45%	How much of the crystal output is fed back. High values build long, growing tails; the loop is soft-clipped to stay controlled.
Damp	0 to 100% · 45%	High-frequency damping in the loop. Higher values give darker, softer tails.
Spread	0 to 100% · 60%	Stereo spread of the crystal layer.
Reverse Chance	0 to 100% · 0%	Probability that a grain plays backwards.
Crystal Only	Off, On · Off	Turns off the filter path so only the crystal layer is heard.

06 Mix and output

Control	Range · default	What it does
Mix	0 to 100% · 100%	Balance between the dry signal and the processed signal.
Low Cut	Off, On · Off	Keeps the low end out of the processed signal.
Output	−18 to +6 dB · 0 dB	Final output level.

High Feedback with long Delay and Size can build up a lot of energy and raise peaks and CPU load. The crystal loop has a fixed buffer of about ten seconds.

07 Presets

Choose a preset from the PRESET list, step through factory presets, or use LOAD USER PRESET... to open a .tbjewel file. Save stores the current settings. The list shows CUSTOM after you edit a preset; Undo and Redo step through changes.

Factory presets: **Initialize**, **Harmonic Lantern**, **Amber Drill**, **Frozen Constellation**, **Phase Dust**, **Reverse Shards**, **Octave Rain**, **Cathedral Shimmer**, and **Fifth Halo**.

08 Recipes

Starting points for common jobs. Adjust them by ear.

In-key shimmer on a pad

A glowing octave layer that follows the harmony.

Preset Cathedral Shimmer

Scale / Base Note Match the song key

Pitch +12 st

Crystal Mix 30 to 50%

- 1 Raise Feedback slowly until the tail sustains between chords.

Arpeggiated resonances

Rhythmic, pitched sparkle from a sustained sound.

Motion Arp Up or Arp Down

Sync On, Division 1/8 or 1/16

Q 8 to 16

- 1 Lower Density for clearer, more separate notes.

Fifth-interval halo

An open, harmonically rich layer above guitars or piano.

Preset Fifth Halo

Pitch +7 st

Spread 60 to 80%

- 1 Check that the layer stays in key with the chords.

Reverse crystal texture

Backwards, glassy grains for ambience and transitions.

Preset Reverse Shards

Reverse Chance 40 to 80%

Crystal Only On

- 1 Blend back with Mix if the texture should sit under the source.

09 Tips and troubleshooting

Symptom	What to check
The shimmer clashes with the music.	Set Base Note and Scale to the key of the song, or use Still motion.
The low end gets muddy.	Switch on Low Cut, or raise the Base Note.
The tail keeps growing.	Lower Feedback, shorten Delay, or raise Damp.
The filter sounds too thin or whistling.	Lower Q or Amount, or switch from Dig to Find.
Random has no effect.	It is used only by Motion types that include randomness.
Synced motion does not follow the song.	Make sure Sync is on and the host is playing; tempos outside 20–320 BPM are clamped.

10 Parameter reference

Panel label	Host parameter	Range	Default
FILTERING			
Base Note	Base Note	MIDI 12 to 108 (C0–C8)	57 (A3)
Scale	Scale	10 scales	Chromatic
Shovel	Shovel	Crystal, Hollow, Bow, Hammer, Metal, Dirt	Crystal
MODE	Filter	Find, Dig	Find
Amount	Amount	0 to 1	0.35
Q	Q	0.35 to 24	6
Density	Density	1 to 24	12
MOTION			
Motion	Motion	10 types	Arp Up
Rate	Rate	0.02 to 64 Hz	0.5 Hz
Sync	Rate Sync	Off, On	Off
Division	Division	1/64 to 1/1	1/4
Motion Depth	Motion Depth	0 to 24 st	7 st
Random	Random	0 to 1	0.5
CRYSTALLIZING			
Crystal Send	Crystal Send	0 to 1	0.8
Crystal Mix	Crystal Mix	0 to 1	0.35
Pitch	Pitch	–24 to +24 st	+12 st
Size	Grain Size	15 to 400 ms	120 ms
Attack	Grain Attack	0.01 to 0.75	0.35
Decay	Grain Decay	0.01 to 0.95	0.65
Delay	Grain Delay	0 to 1000 ms	140 ms
Feedback	Grain Feedback	0 to 100%	45%
Damp	Crystal Damp	0 to 1	0.45
Spread	Crystal Spread	0 to 1	0.6
Reverse Chance	Reverse Chance	0 to 100%	0%
Crystal Only	Crystal Only	Off, On	Off
MIX AND OUTPUT			
Mix	Mix	0 to 1	1
Low Cut	Base Low Cut	Off, On	Off
Output	Output	–18 to +6 dB	0 dB

11 Version history

Version	Changes
1.3.0 · 2026-07-30	The plug-in logo and name share one rack header. Previous/next navigation and save/load in the preset control were improved.

12 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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